

Proof (continued)

We want to show Z is the only nontrivial proper normal subgroup of G .

Let H be a nontrivial normal subgroup of G . First if $H \subseteq B$, then

$$H \subseteq \bigcap_{g \in G} gBg^{-1} = Z \text{ by Lemma.}$$

Now say $H \not\subseteq B$. Set $G' = HB$

Then $B \subseteq G'$ a subgroup of G

$$B \sqcup B \subseteq B$$

$\Rightarrow G' = G$ by the Bruhat decomposition. Then

$$G/H = HB/H \cong B/(B \cap H)$$

Since $D(G) = G \Rightarrow$

$$D(B/(B \cap H)) = B/(B \cap H)$$

But B is solvable $\Rightarrow B = B \cap H$
 $\Rightarrow B \subseteq H$.

So H contains B and all its conjugates
 $\Rightarrow H \supseteq \langle U, sUs^{-1} \rangle = G$, by Lemma 1 $\square$

Proof Let $g \in G$. $C_G(g)$ = centralizer of g in G .

1. $g \in \{\pm I_2\} \Rightarrow C_G(g) = G$.
2. $g \in U \setminus \{I_2\} \Rightarrow C_G(g) = ZU = \{\pm I_2\} \times U$
3. $g \in T \setminus \{\pm I_2\} \Rightarrow C_G(g) = T$
4. $g \in T' \setminus \{\pm I_2\} \Rightarrow C_G(g) = T'$

Proof of 4 Using $d': GL_{\mathbb{F}_q}(\mathbb{F}_{q^2}) \xrightarrow{\sim} GL_2(\mathbb{F}_q)$
 it suffices to show that for $\alpha \in \mathbb{F}_{q^2}^\times \setminus \mathbb{F}_q^\times$

then $C_{GL_{\mathbb{F}_q}(\mathbb{F}_{q^2})}(\alpha) = \mathbb{F}_{q^2}^\times$

Take $h \in C_{GL_{\mathbb{F}_q}(\mathbb{F}_{q^2})}(\alpha)$ and set

$$\beta = h(1) \in \mathbb{F}_{q^2}$$

By assumption

$$h(\alpha) = h(\alpha \cdot 1) = \alpha \cdot h(1) = \alpha \beta$$

And $\alpha \notin \mathbb{F}_q \Rightarrow \mathbb{F}_{q^2} = \mathbb{F}_q \cdot 1 + \mathbb{F}_q \alpha$, so

So for any $a+b\alpha \in \mathbb{F}_{q^2}$,

$$\begin{aligned} h(a+b\alpha) &= ah(1) + bh(\alpha) \\ &= a\beta + b\alpha\beta \\ &= \beta(a+b\alpha) \end{aligned}$$

$$\Rightarrow h = \beta \in \mathbb{F}_{q^2}^\times \quad \square$$

Cor 1. If $q > 3$, $C_G(T) = T$
and $N_G(T) = N = \langle T, s \rangle$

2. $C_G(T') = T'$ and $N_G(T') = N' = \langle T', s \rangle$

Proof 1 If $q > 3$, $\exists g = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \neq \pm I_2$
in T . Then

$$T \subseteq C_G(T) \subseteq C_G(g) = T$$

by the prev Prop. To show $N_G(T) = N$,
we already know $N \subseteq N_G(T)$. Take
 $h \in N_G(T)$. Then

$$hgh^{-1} = h \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} h^{-1} = \begin{pmatrix} b & 0 \\ 0 & b^{-1} \end{pmatrix}$$

with $b = a$ or $b = a^{-1}$, by considering
eigenvalues.

If $b = a$, then $h \in C_G(g) = T \subseteq N$.

If $b = a^{-1}$, then $sh \in C_G(g) = T$

$$\Rightarrow h \in N.$$

2. Similar using $g \in T'$, $g \neq \pm I_2$. $\square$

For $\alpha, \beta \in \mathbb{F}_{q^2}^\times$, write $\alpha \equiv \beta$ if $\beta = \alpha^{\pm 1}$

Fix $x_0 \in \mathbb{F}_q$ such that x_0 is not a square.

Set $u_+ = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $u_- = \begin{pmatrix} 1 & x_0 \\ 0 & 1 \end{pmatrix}$

Thm Assume $p > 2$. Then G has $q+4$ conjugacy classes. A system of representatives is

$$\{I_2, -I_2\} \sqcup \{u_+, u_-, -u_+, -u_-\}$$

$$\sqcup \left\{ d(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} : a \in (\mu_{q-1} \setminus \{\pm 1\}) / \equiv \right\}$$

$$\sqcup \left\{ d'(\alpha) : \alpha \in (\mu_{q+1} \setminus \{\pm 1\}) / \equiv \right\}$$

Proof By considering eigenvalues and canonical forms, these all define distinct conjugacy classes in $GL_2(\mathbb{F}_q)$, except

$$\{u_+, u_-\} \text{ and } \{-u_+, -u_-\}$$

Assume for a contradiction that u_+ and u_- are conjugate in G . So $\exists h \in G$ such that $h u_+ h^{-1} = u_-$. If $v \in \mathbb{F}_q \begin{pmatrix} 1 \\ 0 \end{pmatrix}$,

then $(u_- - I_2) h v = h (u_+ - I_2) v$ by assumption

$$\Rightarrow h v \in \ker(u_- - I_2) = \mathbb{F}_q \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow h \text{ stabilizes } \mathbb{F}_q \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow h \in B$$

Then $h = \begin{pmatrix} a & b \\ c & a^{-1} \end{pmatrix}$

$$\begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & a^{-1} \end{pmatrix}^{-1} = \begin{pmatrix} 1 & x_0 \\ 0 & 1 \end{pmatrix}$$

$\Rightarrow a^2 = x_0$, a contradiction.

The case of $\{-U_+, -U_-\}$ is the same.

To see that every element in G is conjugate to one of these, we count

$$\sum_{\substack{g \in \text{our set} \\ \text{of reps}}} |\text{conj class of } g| = \sum_{\substack{g \in \text{our set} \\ \text{of reps}}} \frac{|G|}{|C_G(g)|}$$

by prev
Prop $1 + 1 + 4 \frac{(q-1)(q+1)}{2}$

$$+ \frac{q-3}{2} \cdot q(q+1) + \frac{q-1}{2} \cdot (q-1)q$$

$$= \dots = (q-1)q(q+1) = |G|. \quad \square$$